

Septicemic Plague: A Comprehensive Disease Characteristics Report

Disease: Septicemic Plague · **MONDO ID:** MONDO:0005956 · **Category:** Infectious Disease
Causative agent: *Yersinia pestis* (NCBI:txid632) · **ICD-10:** A20.7 · **ICD-11:** 1B93.1 · **MeSH:** D010931 (Plague)

Summary

Septicemic plague is an acute, fulminant bloodstream infection caused by the gram-negative coccobacillus *Yersinia pestis*. It is the **least common but most lethal** of the three classical forms of plague. In a systematic review of 762 treated human cases (1937–2019), septicemic plague accounted for only **5% of cases** but carried a **case-fatality rate (CFR) of 38%**, exceeding both bubonic (17%) and pneumonic (27%) forms; untreated, it approaches **~100% fatality** ([PMID: 32435802](#), [PMID: 25643450](#)). It can arise as **primary** septicemic plague — fulminant gram-negative sepsis without a palpable bubo, often after handling infected animal tissue — or as **secondary** septicemic plague when untreated bubonic or pneumonic disease disseminates hematogenously.

The pathophysiology is a coordinated, multi-layered virulence program that first paralyzes innate immunity to permit unchecked bacterial replication, then triggers a destructive inflammatory collapse. Temperature-dependent remodeling of LPS lipid A (from hexa-acylated at the flea's 27 °C to poorly-stimulatory tetra-acylated at the host's 37 °C) blunts TLR4 sensing; a plasmid-encoded **type III secretion system (T3SS)** injects **Yop effectors** that block phagocytosis and cytokine production; the **F1/Caf1 capsule** resists macrophage engulfment; and the **Pla plasminogen activator** enables dissemination from peripheral tissues. This "pre-inflammatory" phase of silent growth is followed by a "pro-inflammatory" phase — a delayed neutrophil influx and cytokine storm (IL-6, TNF- α , IFN- γ) that fails to control bacteria while driving disseminated intravascular coagulation (DIC), acral gangrene, septic shock, and multi-organ failure.

Survival is overwhelmingly determined by **time to effective antibiotics**. Among 533 US patients (1942–2018), mortality was **9% with high-efficacy therapy versus 51% with only limited-efficacy therapy** (PMID: 32435801). There is **no heritable genetic cause**, no established human susceptibility gene, and **no licensed vaccine** for general use, so prevention rests on rodent/flea vector control, exposure avoidance, and post-exposure chemoprophylaxis. This report synthesizes 14 confirmed findings drawn from 42 reviewed papers across all 15 requested sections.

1. Disease Information

Overview. Septicemic plague is the bloodstream-invasive manifestation of infection by *Yersinia pestis*, a gram-negative, non-motile, facultatively intracellular coccobacillus of the family *Enterobacteriaceae*. It is defined clinically by isolation of *Y. pestis* from blood in a patient **without lymphadenopathy (no bubo)** — distinguishing primary septicemic plague from bubonic plague — or by hematogenous dissemination of another plague form (secondary septicemic plague) (PMID: 16943764). The recognized clinical forms of plague are subclinical/serologic plague, plague pharyngitis, *pestis minor* (abortive bubonic), bubonic, **septicemic**, pneumonic, and plague meningitis (PMID: 9097371).

"Clinical presentations include subclinical plague (positive serology without disease); plague pharyngitis; pestis minor (abortive bubonic plague); bubonic plague; septicemic plague; pneumonic plague; and plague meningitis." — PMID: 9097371

Key identifiers.

Resource	Identifier
MONDO	MONDO:0005956 (septicemic plague)
MeSH	D010931 (Plague)
ICD-10	A20.7 (Septicaemic plague); A20 (Plague)
ICD-11	1B93.1 (Septicaemic plague)
NCBI Taxonomy (pathogen)	txid632 (<i>Yersinia pestis</i>)
OMIM	Not applicable (infectious, non-Mendelian)
Orphanet	Plague is not a classical rare-genetic Orphanet entity

Synonyms / alternative names. Septicaemic plague; blood plague; *Yersinia pestis* septicemia; historically part of the "Black Death" (the acral gangrene of DIC gives the blackened appearance).

Data source type. Information for this report is derived from **aggregated disease-level resources** — systematic reviews, case series, surveillance datasets (CDC/WHO), and experimental animal-model studies — rather than individual EHR data.

2. Etiology

Causal factor — infectious only. The sole cause is infection with *Yersinia pestis*. There is **no Mendelian/genetic causation** and no established human susceptibility gene; the disease is **not heritable** (inheritance not applicable) ([PMID: 9097371](#)).

Risk factors (environmental/exposure). Documented factors include residence in or travel to endemic foci; occupational and recreational animal contact (veterinarians and assistants, hunters/trappers, wildlife biologists); pet ownership and direct animal-reservoir contact (especially during hunting season); living in a household with an index case; flea bite; **handling infected animal tissue** (a route that can produce primary septicemic plague); and climatic conditions favoring reservoir/vector abundance (mild winters, cool moist springs, early summers) ([PMID: 9097371](#)). Cats are a notable source of respiratory (pneumonic) transmission to humans.

"Other factors that increase risk of infection in endemic areas are occupation-veterinarians and assistants, pet ownership, direct animal-reservoir contact especially during the hunting season, living in households with an index case, and, mild winters, cool moist springs, and early summers." — PMID: 9097371

Immunocompromise. HIV and other immunocompromising conditions can worsen severity/outcome; a 2024 Nigerian case report highlights the potential severity of co-infection in immunocompromised individuals (PMID: 41263683).

Genetic protective factors / gene–environment interactions. No validated human protective alleles or GxE interactions are established for plague susceptibility. (Historical hypotheses linking *CCR5-Δ32* or other loci to plague survival remain unproven.)

3. Phenotypes

Septicemic plague presents as **fulminant gram-negative sepsis without a palpable bubo**: high fever, chills, rigors, prostration, tachycardia, hypotension, and rapid progression to shock, DIC, and multi-organ failure. **Acral necrosis/gangrene** of digits and nose is a classic late sign (the basis of "Black Death") (PMID: 16943764, PMID: 9097371).

"Five (38%) patients had primary septicemic plague, and the remaining eight (62%) had bubonic plague." — PMID: 16943764

Phenotype	Type	Onset / progression	Frequency	Suggested HPO
Fever, chills	Symptom/sign	Acute (1–7 d incubation), rapidly progressive	Near-universal	HP:0001945 (Fever)
Hypotension / septic shock	Clinical sign	Rapid	Common in severe/late disease	HP:0001635 hypotension; HP:0031273 septic shock
Disseminated intravascular coagulation	Lab/clinical	Rapid	Characteristic of severe disease	HP:0005521 (DIC)
Acral gangrene / necrosis (digits, nose)	Physical manifestation	Late	Classic but not universal	HP:0100758 (Gangrene)
Abdominal pain, nausea, vomiting, diarrhea	Symptom	Early/variable	Frequent in septicemic form	HP:0002027; HP:0002018; HP:0002014
Prostration / altered mental status	Symptom/sign	Progressive	Common	HP:0012378 (Fatigue); HP:0001259 (Coma, severe)
Thrombocytopenia, leukocytosis	Lab abnormality	Early	Frequent	HP:0001873; HP:0001974
Absence of bubo	Distinguishing feature	—	Defining for <i>primary</i> septicemic	—

Severity/progression. Severe and progressive; without early antibiotics, the course is fulminant and lethal within days. **Quality-of-life impact:** acute and life-threatening rather than chronic; survivors of DIC-associated gangrene may require amputation with long-term disability, but there is no chronic/relapsing phase.

4. Genetic / Molecular Information

Not applicable in the human-host sense. Septicemic plague has **no causal human genes, no pathogenic germline/somatic variants, no modifier genes, no disease-specific epigenetic signature, and no chromosomal abnormalities** — it is a purely infectious disease.

The relevant "genetics" are those of the **pathogen**, whose plasmid-encoded virulence genes drive the septicemic phenotype:

Genetic element	Location	Product / function
<i>pla</i>	pPCP1 (pKYP1, ~9.5 kb)	Plasminogen activator (Pla): coagulase + fibrinolysin; dissemination (PMID: 8360901)
T3SS + <i>yop</i> effectors (<i>yopH</i> , <i>E</i> , <i>OypkA</i> , <i>M</i> , <i>J</i> , <i>T</i>), <i>lcrV</i>	pCD1 (pYV, ~70 kb)	Type III secretion; injection of anti-immune effectors (PMID: 15847602 , PMID: 24599533)
<i>caf1M1A1</i> operon	pMT1 (~100 kb)	F1 (Caf1) capsular antigen; antiphagocytic (PMID: 35358289 , PMID: 19103769)
lipid A biosynthesis genes	chromosome	Temperature-dependent lipid A acylation; TLR4 evasion (PMID: 23745121)

Y. pestis is a recently emerged, genetically monomorphic clone of *Y. pseudotuberculosis*. Rare naturally occurring **F1-negative (caf-negative)** strains remain virulent but evade F1-based diagnostics ([PMID: 35320275](#)).

5. Environmental Information

- **Infectious agent:** *Yersinia pestis* (NCBI:txid632), gram-negative coccobacillus, *Enterobacteriaceae*.
- **Environmental/ecological factors:** maintenance in sylvatic rodent–flea cycles; climatic conditions promoting reservoir and flea abundance (mild winters, cool moist springs) increase spillover risk ([PMID: 9097371](#)).
- **Occupational/lifestyle exposures:** hunting/trapping, veterinary work, wildlife biology, handling animal carcasses; the tissue-handling route is specifically linked to primary septicemic plague.

- No chemical toxin, radiation, smoking, diet, or pollution etiology applies.

6. Mechanism / Pathophysiology

Ordered causal chain

1. *Y. pestis* enters the host by **flea bite** or by **contact with infected tissue/aerosol** → deposits bacteria in skin/lymphatics or bloodstream.
2. On shifting from 27 °C (flea) to 37 °C (host), the bacterium **remodels LPS lipid A** to a tetra-acylated, poorly-stimulatory form → **evades TLR4/MD-2 recognition** (PMID: 23745121) → *results in* delayed innate alarm.
3. The bacterium assembles the **T3SS** and injects **Yop effectors** into macrophages (and later neutrophils): YopE inhibits phagocytosis; YopJ induces macrophage cytotoxicity/apoptosis and blocks proinflammatory cytokines; Yops disrupt cytoskeletal dynamics → *leads to* failure of early innate clearance and **extracellular multiplication in lymphoid tissue** (PMID: 15847602, PMID: 24599533).
4. The **F1/Caf1 capsule** coats the bacterium, resisting macrophage engulfment (polymer-brush effect, ~400 pN mechanical stability) → *reinforces* immune evasion (PMID: 35358289).
5. Yop-mediated suppression of **LTB4 synthesis** delays neutrophil/macrophage inflammation → *prolongs* the silent growth window (PMID: 39423229).
6. The **Pla plasminogen activator** (coagulase/fibrinolysin) degrades fibrin barriers → *enables* dissemination from the peripheral site into blood (PMID: 8360901).
7. Bacteria disseminate hematogenously — in animal models, **spleen → liver → blood** — reaching high bacteremia (PMID: 19073275).
8. **Branch — pre-inflammatory → pro-inflammatory switch**: after ~48 h the host mounts a **delayed neutrophil influx and cytokine storm** (IL-6, TNF- α , IFN- γ , IL-12p70, MCP-1, KC, MIP-2) that fails to control bacteria but *causes* tissue destruction (PMID: 24098126, PMID: 17101642).
9. Systemic inflammation + coagulopathy → **DIC**, microvascular thrombosis, **acral gangrene**, **septic shock**, **ARDS**, and **multi-organ failure** → *results in* death within days if untreated (PMID: 16943764, PMID: 34780267).

Biphasic model (schematic)

PHASE 1 – IMMUNE EVASION (pre-inflammatory)
 [37°C lipid A shift -> TLR4 blind]
 [T3SS/Yops -> block phagocytosis + cytokines] ==>
 [F1 capsule -> antiphagocytic]
 [Pla -> dissemination]
 silent bacterial growth spleen->liver->blood

PHASE 2 – CYTOKINE STORM (pro-inflammatory)
 [~48 h: neutrophil influx]
 [IL-6, TNF- α , IFN- γ surge]
 [tissue destruction, DIC]
 [septic shock, MOF, gangrene]
 host collapse

"Pneumonic plague progression is biphasic, with an initial pre-inflammatory phase facilitating bacterial growth in the absence of host inflammation, followed by a pro-inflammatory phase marked by extensive neutrophil influx, an inflammatory cytokine storm, and severe tissue destruction." — [PMID: 24098126](#)

"Effector Yops function to counteract multiple signaling responses in the infected host cell ... Innate and adaptive immune responses are thwarted as a consequence of Yop activities." — [PMID: 15847602](#)

Upstream vs downstream. Upstream = LPS remodeling, T3SS/Yops, F1 capsule, Pla (immune evasion + dissemination). Downstream = cytokine storm, DIC, shock, organ failure. Blunting the downstream inflammation (intranasal fluticasone) reduced IL-6, neutrophil infiltration, bacterial burden, and improved antibiotic-treated survival — evidence the storm itself is pathogenic ([PMID: 34780267](#)).

Suggested ontology terms. GO:0006955 (immune response), GO:0006954 (inflammatory response), GO:0042742 (defense response to bacterium), GO:0030193 (regulation of blood coagulation), GO:0006909 (phagocytosis). Cell types: CL:0000235 (macrophage), CL:0000775 (neutrophil), CL:0000115 (endothelial cell). CHEBI:16412 (lipopolysaccharide).

7. Anatomical Structures Affected

- **Primary/secondary organs (organ level): blood/vasculature** (primary site of septicemic disease, UBERON:0000178 blood; UBERON:0001981 blood vessel), **spleen** (UBERON:0002106) and **liver** (UBERON:0002107) as major dissemination targets ([PMID: 19073275](#)); **lymph nodes** (UBERON:0000029) in secondary-from-bubonic cases; **lungs** (UBERON:0002048) in

secondary pneumonic progression; **skin/acral tissue** (UBERON:0002097) in DIC gangrene; **brain/meninges** in plague meningitis (UBERON:0002360).

- **Body systems:** cardiovascular (shock, DIC), hematologic/lymphoid, hepatobiliary, respiratory (ARDS), and nervous (meningitis) systems.
- **Tissue/cell level:** vascular endothelium; reticuloendothelial macrophages (CL:0000235); neutrophils (CL:0000775). *Y. pestis* multiplies **extracellularly** in lymphoid tissue after initial macrophage interaction.
- **Subcellular level:** T3SS delivers effectors into the host **cytoplasm** (GO:0005737); F1 acts at the bacterial **cell surface/extracellular region** (GO:0005576); LPS resides in the **outer membrane** (GO:0009279).
- **Lateralization:** systemic/bilateral; acral gangrene is typically symmetric-peripheral.

8. Temporal Development

- **Onset:** acute, after a **1–7 day incubation** (often shorter — hours to a few days — for septicemic/pneumonic forms) ([PMID: 9097371](#)).
- **Progression:** rapid and fulminant; untreated septicemic plague progresses to death within days (pneumonic ~6 days in models; septicemic similarly rapid). For late-stage disease, **antibiotics must be given within ~24 h of symptom onset** to be effective ([PMID: 34780267](#)).
- **Course pattern:** monophasic and progressive — **no chronic, relapsing, or remitting phase**; outcome is either death or recovery.
- **Critical period:** the narrow **early therapeutic window** is the dominant modifiable variable (see Section 11).

"Late-stage pneumonic plague is difficult to treat, as antibiotics must be delivered within 24 h after onset of symptoms to be effective." — [PMID: 34780267](#)

9. Inheritance and Population

Epidemiology. Between 2010 and 2019 the six countries reporting the most human plague cases were **Madagascar, DR Congo, Uganda, Peru, Tanzania, and the USA**, totaling **4,547 cases with 17% mortality (786 deaths)** ([PMID: 40022523](#)). *Y. pestis* is a **WHO priority pathogen with epidemic/pandemic potential**, endemic in rodent reservoirs across Africa, Asia, North America, and South America; **Madagascar** records a large share of annual global cases ([PMID: 33264458](#), [PMID: 38935608](#)). Septicemic plague constitutes ~5% of plague cases ([PMID: 32435802](#)).

"Between 2010 and 2019, the six countries with the most reported human cases of Yersinia pestis infection ... were Madagascar, the Congo, Uganda, Peru, Tanzania, and the USA, with a total of 4,547 cases with a mortality rate of 17% (786 cases)." — [PMID: 40022523](#)

Genetic etiology. Not applicable — no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency; the disease is infectious.

Population demographics. Risk tracks **exposure**, not ethnicity — rural residents, hunters, and animal handlers in endemic foci are over-represented. Geographic distribution follows enzootic rodent foci (western US, Madagascar, central Asia, Andean South America, sub-Saharan Africa). No strong intrinsic sex bias independent of occupational exposure; all ages are susceptible.

10. Diagnostics

Confirmatory diagnosis rests on three pillars — **culture, PCR, and serology** — supplemented by rapid antigen tests. For the **septicemic form specifically, blood culture and peripheral blood smear** (bipolar "safety-pin" gram-negative coccobacilli on Wright/Giemsa/Wayson stain) are central because there is no bubo to aspirate.

Method	Target/approach	Performance
Bacterial culture	Isolate <i>Y. pestis</i> from blood/aspirate/sputum (reference standard)	~65% sensitivity
PCR / qPCR	<i>caf1, pla, yopM</i>	~85% (single); triplex qPCR 100% sens / 82% spec (PMID: 40705833)
Serology	Anti-F1 IgG	positive in ~93% of confirmed cases
F1 rapid diagnostic test (F1RDT / LFI)	Capsular F1 antigen, point-of-care	On-site 94% sens / 74% spec (PMID: 41389991); Madagascar retrospective 100% sens / 67% spec (PMID: 32000692)
Dual-antigen (F1+LcrV) LFI/ELISA	Detects F1-negative strains too	LoD ~1–2 ng/mL (PMID: 35320275)
ddPCR (multi-target)	<i>ypo2088, caf1, pla</i>	Superior sensitivity for low-load samples (PMID: 38701065)

"The sensitivity and specificity of on-site F1RDT were 94% (95% CI, 89.6-97.0) and 74% (95% CI, 68.2-79.3) against RS1" — PMID: 41389991

Genetic/omics testing: not applicable to human diagnosis; pathogen genotyping (MLVA, CRISPR, SNP/WGS) is used for outbreak epidemiology, not patient diagnosis (PMID: 38935608).

Differential diagnosis: other causes of gram-negative sepsis/DIC, meningococemia, tularemia, anthrax, rickettsioses, and other hemorrhagic febrile illnesses; the epidemiologic exposure history and blood smear morphology are key discriminators. **Blind spot:** rare F1-negative strains evade F1-based antigen tests — hence dual-antigen assays and PCR backup.

11. Outcome / Prognosis

Mortality. Treated septicemic plague CFR is ~38%; untreated it approaches ~100% (PMID: 32435802, PMID: 25643450). The **single dominant prognostic factor is time to effective antimicrobial therapy**: in 533 US patients, mortality was 9% with high-efficacy therapy vs 51% with limited-efficacy therapy (PMID: 32435801).

"Without antibacterial therapy, the disease is associated with a high case fatality rate, ranging from 40% (bubonic plague) to nearly 100% (septicemic and pneumonic plague)." — PMID: 25643450

"Mortality differed significantly among those receiving high-efficacy therapy (9%) and only limited-efficacy therapy (51%)." — PMID: 32435801

Complications. Septic shock, **DIC with purpura and acral gangrene** (fingers, toes, nose), ARDS, secondary pneumonic plague (with human-to-human transmission risk), plague meningitis, and multi-organ failure (PMID: 34780267).

Recovery. Binary — death or recovery, with no chronic phase. Survivors of DIC-associated gangrene may require **amputation**, producing long-term functional disability. Other prognostic factors: delay in diagnosis, older age, immunocompromise (e.g., HIV, PMID: 41263683), and severity of shock/DIC at presentation.

12. Treatment

Pharmacotherapy is the mainstay and must be started empirically on clinical suspicion — before laboratory confirmation.

Drug class	Agents	Notes / NCIT
Aminoglycosides (first-line)	Streptomycin, gentamicin	Aminoglycoside-treated CFR ~13% vs 20% overall; streptomycin outperformed gentamicin in US data (PMID: 32435801). NCIT:C540 (Gentamicin), NCIT:C839 (Streptomycin)
Fluoroquinolones	Ciprofloxacin, levofloxacin	FDA-approved for plague; used in >30% of recent patients. NCIT:C2249 (Ciprofloxacin)
Tetracyclines	Doxycycline	Associated with increased survival; oral option. NCIT:C516 (Doxycycline)
Others	Chloramphenicol, sulfonamides/TMP-SMX	Chloramphenicol preferred historically for plague meningitis (CNS penetration)

"Gentamicin use was associated with higher mortality than streptomycin, and aminoglycoside use was linked to higher mortality than for tetracyclines." — [PMID: 32435801](#)

Intracellular caveat: during the early facultative-intracellular stage, streptomycin and ciprofloxacin retain efficacy against intracellular *Y. pestis*, whereas gentamicin and doxycycline are less effective intracellularly — relevant for agent selection ([PMID: 21628541](#)).

Supportive care: aggressive management of septic shock, fluid resuscitation, vasopressors, and DIC/coagulopathy management are essential given the fulminant sepsis physiology.

Adjunctive anti-inflammatory strategies are experimental — fluticasone improved antibiotic-treated survival in a pneumonic model by dampening the cytokine storm ([PMID: 34780267](#)).

Surgery: amputation/debridement of gangrenous tissue in survivors. No gene, cell, RNA, or targeted/immuno-therapies apply. No pharmacogenomic considerations are established.

13. Prevention

- **Primary prevention:** rodent and flea **vector control**, insecticides, environmental sanitation, avoiding contact with sick/dead animals, and personal protective equipment when handling potentially infected tissue ([PMID: 32853251](#)).

- **Post-exposure prophylaxis (secondary): doxycycline or ciprofloxacin for 7 days** for close contacts of pneumonic cases; isolation of pneumonic patients for the first ~48 h of therapy prevents human-to-human spread (PMID: 29183475).
- **Immunization:** killed whole-cell and live-attenuated vaccines are no longer used (safety/efficacy). Modern subunit candidates combine the two protective antigens **LcrV** (T3SS needle-tip) and **F1** (capsule) — rF1+rV, rF1V fusion, rV10. Antibodies against LcrV and F1 interfere with T3SS injection of host cells (PMID: 19786842). As of 2024, **>20 candidates are in preclinical development, few in phase 1, and none licensed for general use** (PMID: 40022523).
- **No genetic screening/counseling** applies (non-heritable). Public-health surveillance and rapid outbreak response remain the backbone of control.

"LcrV, a protein at the tip of type III secretion needles, and F1, the capsular pilus antigen, are both recognized as plague protective antigens. Antibodies against LcrV and F1 interfere with Y. pestis type III injection of host cells." — PMID: 19786842

"More than 20 candidate plague vaccines are in the preclinical phase, with few in early (phase 1) clinical trials." — PMID: 40022523

14. Other Species / Natural Disease

- **Taxonomy of hosts:** *Y. pestis* infects **>200 mammal species**. Keystone reservoirs include ground squirrels, prairie dogs, marmots, gerbils, and rats (*Rattus* spp.). Fleas — especially *Xenopsylla cheopis* — are the vectors (PMID: 33264458).
- **Transmission / zoonosis:** maintained in **sylvatic rodent–flea cycles**; fleas transmit after Pla/biofilm-mediated proventricular blockage. Humans are **incidental hosts** infected by flea bite, handling infected animals, or respiratory spread (notably from cats) (PMID: 8360901, PMID: 9097371).
- **Natural disease / veterinary relevance:** cats develop severe (often pneumonic) plague and are a documented source of human infection; wild rodents and lagomorphs suffer epizootic die-offs.

- **Comparative biology:** *Y. pestis* is an evolutionarily recent clone of *Y. pseudotuberculosis*; the disease mechanisms (T3SS, F1) are conserved across mammalian hosts, underpinning the utility of animal models.

15. Model Organisms

- **Standard models:** **mouse** (*Mus musculus*, NCBI:txid10090) and **rat** (*Rattus norvegicus*, NCBI:txid10116), typically challenged with the fully virulent strain **CO92** by intranasal/aerosol or intradermal routes ([PMID: 19073275](#), [PMID: 31177429](#)).
- **Phenotype recapitulation:** models reproduce the **dissemination pattern relevant to septicemic spread** — bacteria move from the primary site to spleen, then liver and blood, with rising serum cytokines/chemokines and histopathologic injury over 72 h. Rats are **as susceptible as mice** (similar LD50) and support **animal-to-animal transmission** ([PMID: 19073275](#)).

"Bacteria disseminated from the lungs to peripheral organs, with the largest increases in the spleen, followed by the liver and blood at 72h p.i." — [PMID: 19073275](#)

"rats were as sensitive to pneumonic plague as mice, having a similar LD(50) dose by the intranasal and aerosolized routes. Further, we showed direct transmission of plague bacteria from infected to uninfected rats." — [PMID: 19073275](#)

- **Non-human primates** are used for vaccine efficacy testing under the FDA "Animal Rule."
 - **Limitations:** rodent inflammatory kinetics differ from humans; anesthesia and intranasal instillation add variability (hence standardized aerosol challenge); models best capture pneumonic/systemic pathophysiology rather than the specific human primary-septicemic tissue-handling route.
-

Mechanistic Model / Interpretation

Septicemic plague is best understood as a **two-act drama of immune subversion followed by immune catastrophe**. In Act I, *Y. pestis* deploys a layered virulence toolkit — temperature-tuned low-TLR4 lipid A, T3SS-delivered Yop effectors, the F1 antiphagocytic capsule, and Pla-driven dissemination — that renders the early innate response blind and impotent, allowing silent exponential growth and hematogenous spread (spleen → liver → blood). In Act II, once bacterial burden is overwhelming, a **delayed and futile neutrophil/cytokine storm** erupts; it fails to clear the pathogen but ignites DIC, microvascular thrombosis, acral gangrene, septic shock, and multi-organ failure. This framework explains the two clinical hallmarks of the septicemic form — its **fulminant tempo** (silent growth means patients present already deep into dissemination) and its **exceptional lethality** (immune paralysis then immune-mediated tissue destruction) — and it explains why **time-to-antibiotic is decisive**: therapy given before the storm interrupts the cascade, while therapy given after it (>24 h in late disease) cannot reverse established DIC/shock. The finding that anti-inflammatory adjuncts improve antibiotic-treated survival further supports the storm as an independently pathogenic downstream node, and points to combined antimicrobial + immunomodulatory strategies as a rational future direction.

Evidence Base

PMID	Contribution
32435802	Septicemic plague = 5% of cases, 38% CFR (systematic review, 762 cases)
25643450	Near-100% untreated fatality; <i>Y. pestis</i> etiology
40022523	2010–2019 global epidemiology; vaccine pipeline status
33264458	Rodent-reservoir endemicity; Madagascar hotspot
8360901	<i>pla</i> coagulase/fibrinolysin → dissemination/transmission
23745121	Temperature-dependent lipid A → TLR4 evasion
15847602	Yop effectors thwart innate + adaptive immunity
24599533	YopE (anti-phagocytosis), YopJ (cytotoxicity); LcrV antigen
39423229	Yop-mediated LTB4 suppression delays inflammation
35358289	F1/Caf1 antiphagocytic capsule
19103769	Anti-F1 antibodies protective; capsule operon in transmission
16943764	Primary septicemic plague clinical cluster (US, 2006)
9097371	Clinical forms; risk factors
32435801	Therapy efficacy: 9% vs 51% mortality; agent comparisons
24098126	Biphasic pre-/pro-inflammatory mechanism; macrophage→neutrophil switch
17101642	Delayed (~48 h) cytokine/chemokine surge; named mediators
19073275	Rat model; spleen→liver→blood dissemination; transmission
34780267	24 h therapeutic window; fluticasone adjunct benefit
41389991 / 32000692 / 40705833 / 35320275	Diagnostics: F1RDT, triplex qPCR, dual-antigen LFI
19786842	LcrV + F1 subunit vaccine antigens
21628541	Intracellular antibiotic efficacy differences
29183475	Prophylaxis and isolation guidelines

PMID	Contribution
41263683	Fulminant plague in HIV-positive patient

Evidence source types: human clinical/surveillance (systematic reviews, case series, RDT field studies), model organism (mouse/rat CO92 studies), and in vitro/molecular (T3SS, F1, Pla, lipid A mechanism papers).

Limitations and Knowledge Gaps

1. **Septicemic-specific data are sparse.** Much mechanistic evidence (biphasic model, cytokine storm) derives from **pneumonic** models; the extent to which these kinetics apply to primary septicemic plague is inferred, not directly demonstrated.
2. **No human genetic/host-susceptibility data.** Whether host polymorphisms modulate septicemic risk or outcome is essentially unstudied; historical protective-allele hypotheses remain unproven.
3. **Comparative antibiotic effectiveness** derives largely from observational US data (confounding by indication); randomized head-to-head trials in septicemic plague are lacking.
4. **Diagnostic blind spots.** F1-based rapid tests miss rare F1-negative strains; blood-culture sensitivity (~65%) and turnaround limit rapid confirmation in fulminant sepsis.
5. **No licensed vaccine;** correlates of protection in humans are incompletely defined.
6. **Adjunctive immunomodulation** is supported only by animal data.

Proposed Follow-up Experiments / Actions

1. **Dedicated septicemic-plague models:** intravenous/intradermal CO92 challenge with serial organ bacteriology and single-cell/transcriptomic profiling to confirm the biphasic evasion→storm cascade specifically for the septicemic route.
2. **Adjunctive anti-inflammatory trials:** evaluate corticosteroids or targeted cytokine blockade (anti-IL-6/anti-TNF) plus antibiotics in animal septicemic models, building on the fluticasone result ([PMID: 34780267](#)).

3. **Next-generation diagnostics:** deploy dual-antigen (F1+LcrV) and multi-target PCR/ddPCR point-of-care assays to close the F1-negative blind spot and shorten time-to-diagnosis in bacteremic patients ([PMID: 35320275](#), [PMID: 38701065](#)).
 4. **Host-genetics study:** GWAS/exome analysis of survivors vs fatal cases in endemic foci (e.g., Madagascar) to test for host modifiers of outcome.
 5. **Vaccine advancement:** move optimized rF1V/rV10 subunit candidates through phase 1/2, defining human immune correlates ([PMID: 19786842](#), [PMID: 40022523](#)).
 6. **Antibiotic-selection optimization:** prospective comparative-effectiveness data (streptomycin vs gentamicin vs fluoroquinolone vs tetracycline), incorporating intracellular-efficacy considerations ([PMID: 32435801](#), [PMID: 21628541](#)).
-

Report compiled from 14 confirmed findings across 42 reviewed papers, covering all 15 requested disease-characteristic sections. Ontology suggestions (MONDO, HPO, GO, CL, UBERON, CHEBI, NCIT, NCBI Taxonomy) are provided inline where applicable.
